

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD) "Coding Requirements" section:

FR-DLT-001: Load Customer and Order Data from CSV to Delta Tables

Title: Load Customer and Order Data from CSV to Delta Tables

Description: The system must read customer and order data from CSV files and load it into delta tables, applying data cleansing and transformation rules.

Preconditions:

- CSV files are available at the specified locations (/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata and /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata)
- Delta tables "customer" and "order" exist or are created

Main Flow / Functional Steps:

- Read customer data from CSV file (/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata) into a delta table "customer"
- Read order data from CSV file (/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata) into a delta table "order"
- Apply schema to "customer" and "order" delta tables (CustId, Name, EmailId, Region for "customer" and OrderId, ItemName, PricePerUnit, Qty, Date, CustId for "order")
- Add a new column "TotalAmount" to the "order" delta table, calculated as the product of "PricePerUnit" and "Qty"

FR-DLT-002: Cleanse Customer and Order Data

Title: Cleanse Customer and Order Data

Description: The system must remove null and duplicate records from the "customer" and "order" delta tables.

Preconditions:

- "customer" and "order" delta tables exist and contain data

Main Flow / Functional Steps:

- Remove null records from "customer" and "order" delta tables
- Remove duplicate records from "customer" and "order" delta tables

FR-DLT-003: Create ordersummary Table and Load Data

Title: Create ordersummary Table and Load Data

Description: The system must create the "ordersummary" table if it does not exist and load data into it by joining "customer" and "order" delta tables.

Preconditions:

- "customer" and "order" delta tables exist and contain cleansed data
- "ordersummary" table does not exist or is recreated

Main Flow / Functional Steps:

- Create "ordersummary" table if it does not exist in catalog="gen_ai_poc_databrickscoe" and schema="sdlc_wizard"
- Join "customer" and "order" delta tables on "CustId" and load the joined data into "ordersummary" table
- Apply SCD type 2 logic to "ordersummary" table

FR-DLT-004: Update ordersummary Table on Customer Changes

Title: Update ordersummary Table on Customer Changes

Description: The system must update the "ordersummary" table when there are changes to the "customer" table, applying SCD type 2 logic.

Preconditions:

- "customer" and "ordersummary" tables exist and contain data

Main Flow / Functional Steps:

- Detect changes to the "customer" table
- Update "ordersummary" table by making old records inactive and new records active, and updating StartDate and EndDate accordingly

FR-DLT-005: Create customeraggregatespend Table and Load Aggregated Data

Title: Create customeraggregatespend Table and Load Aggregated Data

Description: The system must create the "customeraggregatespend" table if it does not exist and load aggregated data into it from the "ordersummary" table.

Preconditions:

- "ordersummary" table exists and contains data
- "customeraggregatespend" table does not exist or is recreated

Main Flow / Functional Steps:

- Create "customeraggregatespend" table if it does not exist in catalog="gen_ai_poc_databrickscoe" and schema="sdlc_wizard"
- Aggregate "TotalAmount" from "ordersummary" table by "Name" and "Date"
- Load aggregated data into "customeraggregatespend" table

FR-DLT-006: Implement Data Processing using Delta Live Tables

Title: Implement Data Processing using Delta Live Tables

Description: The system must implement the data processing pipeline using Delta Live Tables (DLT).

Preconditions:

- All previous requirements are implemented

Main Flow / Functional Steps:

- Implement steps 1 to 5 using DLT, ensuring data is processed in the correct order and with the required transformations.